

Tips for Computer Science Faculty Applications

Medium Search Write

Sign up

Sign in



196



2



(Last updated September 28th, 2024.)

This article is a collection of tips for improving your faculty application package (tailored to computer science). This article is going to be a living document and will be updated over time. This article is also biased towards how applications are evaluated at research universities in the US and Canada.

Topics covered include: Cover Letter, Research Statement (most important thing under your control), Teaching Statement, Diversity Statement, Representative Publications, Letters of Recommendations.

There are also many other resources written by other people that covers a broader range including interviewing, etc.: [Michael Ernst](#), [Shomir Wilson](#), [Aaditya Ramdas](#). (to be updated as appropriate)

. . .

Cover Letter

The cover letter is primarily a routing tool to get the application looked at by the right reviewers. A second goal is to highlight any special circumstances in your application. With that in mind, it's good to state things like:

Position you're applying for. This includes rank (assistant, associate, etc.), as well as specific title of the position opening as listed on the job posting.

Your area of expertise. Stating this clearly helps get your application routed to the appropriate reviewer. A mis-routing here can lead to significant delays or even errors in the evaluation.

Whether you're applying confidentially. This point mainly applies to senior applicants.

Whether you want to disclose a 2-body problem. It might make sense to disclose this information if you and your partner are both applying at the same time. The default advice is to not to disclose any 2-body problems until after you receive an offer, but occasionally it does make sense to disclose up front.

Keep it short. The main purpose of a cover letter is for routing purposes. The more information you put in it that is not related to that, the more likely the cover letter is misread.

. . .

Research Statement

The research statement is your primary vehicle for articulating a research vision. This is the only part of the application package for you to do so.

Quick Tips. The committee is trying to judge the significance and intellectual merit of your research agenda. So don't try to sell your accomplishments or skills piece-meal, but rather sell your vision. Moreover, while the research statement is an opportunity to brag, that bragging is best served to provide supporting evidence that you are the best qualified person to carry out the stated agenda.

Reference statements (to be updated):

- [My research statement](#)
- [Sarah Dean's research statement](#)
- [Angie Liu's research statement](#)

- [Guanya Shi's research statement](#)
- [Ziv Scully's research statement](#)
- [Sara Beery's research statement](#)
- [Yuxin Chen's research statement](#)
- [Anjalie Field's research statement](#)
- [Jennifer Sun's research statement](#)

Significance. What is the significance of the research agenda you wish to pursue? In other words, how will your research agenda, if successful, change the world?

In my field of machine learning, some strategies include:

- Identifying *blocking limitations of existing methods* (which requires significant tuning & engineering to get things working).
- Identifying *strategies to help us better understand* those aspects.
- Identifying *new applications are enabled* as a consequence.
- Identifying *novel intellectual connections* between machine learning and other fields, thus bridging two fields together.

Many strong research statements include a combination of the above strategies. The goal is to paint a picture that the whole is greater than the sum of the parts, and that the impact is potentially far reaching. Of course, one very important aspect of significance is that it is intellectually interesting, which deserves its own category (discussed next).

Intellectual Merit. Academic research jobs are ultimately about pursuing intellectually rich research. As such, your research vision should reveal rich intellectual questions (as well as have significance as discussed above). Such intellectual questions should be somehow fundamental and cross-cutting (it's up to you to figure out how to frame it that way). Your prior work is most useful here, as it can be used as evidence of intellectual merit of this research agenda.

In machine learning & related fields, some examples include:

- Developing new theoretical tools that *expands our ability to analyze learning systems* (e.g., [Neural Tangent Kernel](#) & [Sample Complexity of Algorithm Design](#))
- Comprehensively exploring some limitation in machine learning, *revealing*

layers of unexpected implications (e.g., Recognition in Terra Incognita & Human Decisions and Machine Predictions).

- Finding a *deep mathematical property* that enables more effective learning, possibly uncovering connections to other fields (e.g., Score-Based Generative Models & Thompson Sampling with Langevin Dynamics).
- *Rethinking and articulating the relevant principles* of building data-driven systems, which reflect emerging challenges & opportunities (e.g., Data Programming & XGBoost).
- Developing an approach that *enables new capabilities*, and articulate the lessons learned in doing so (e.g., Neural-Control Family & DreamCoder).
- Framing a *new problem* or setting (e.g., Mechanism Design for Social Good & Adaptive A/B Testing via Online FDR Control).
- Building new ways to *systematically encode prior knowledge that removes bottlenecks* such as labeling bottlenecks or aliasing effects (e.g., Task Programming & Mask R-CNN).

Cut the Fluff. Faculty search committees read hundreds of research statements. They don't need to read another generic paragraph about self-driving cars, LLMs that hallucinate, or studying fundamental limits of computation (aka proving matching lower bounds). Aggressively cut the fluff, and get to the point of what makes you unique. ***Based on my experience advising and evaluating faculty applicants, up to 50% of a research statement is fluff (especially a first draft).***

Extrapolate Boldly yet Thoughtfully. Your research vision is about what can be possible, so it's important to have a vision that extends into the future. Your prior work is evidence that this vision has legs, but the future work is place where you are charting a path forward. I think this is the most distinctive piece of information that the research statement provides. A few strategies that I like to use are discussed next.

An Interesting “Medium-Level” Agenda. It's relatively easy to craft a good high-level agenda (e.g., AI for Science, AI for Social Good, Protein Modeling, Real-World Robotics, etc.). It's also pretty easy to get into the details of your research (e.g., summarizing the key findings in individual papers). However, it is the medium-level story that often ties your research agenda together in a coherent and intellectually interesting way. The alternative is to jump straight from a relatively vague/abstract high-level story into what might feel

like a laundry list of projects and results. What are the key insights that propel you forward when you seek out new projects? Why are your previous results realizations or instantiations of these insights? *****Based on my experience advising and evaluating faculty applicants, this is the most important thing to work on.*****

Articulate the Principal Component of your Research (in the First Page).

This point is closely related to the above one. Essentially, people want to know (roughly) what you'll get tenure on. The best way to do this is for people to understand the principal component of your research, because that allows them to extrapolate your research into the future. In having a well-articulated medium-level agenda, you are essentially showing others what your principal component is (otherwise, people will have to construct it themselves from inspecting your papers individually).

Moreover, you should strongly consider making this principal component clear in the first page of your research statement. In fact, you should assume that many readers will not read beyond the first page.

The Principal Component & Medium Level Agenda Should Transcend Your Prior Work. The framing of your research vision should not read like a summary of your prior work. You are being hired for what you want to do in the future. The prior work is simply there to show that your research vision is not smoke & mirrors. Instead, the framing should ideally integrate both your prior work and future work into a coherent new thing.

Get Yisong Yue's stories in your inbox

Join Medium for free to get updates from this writer.

Adversarial Bullshitting. To be blunt, extrapolating boldly will require some amount of projecting a future that might never exist. So go ahead! Of course, one then needs to examine the proposed bold ideas critically to see if they can be made to flow logically from the research vision. One can think of this as an adversarial minimax game of making bold extrapolations (maximization) while criticizing them for being too unrealistic

(minimization). Hopefully, the equilibrium solution is one that strikes the right balance in terms of being visionary yet grounded. Ideally, this equilibrium solution can be distilled into a medium-level agenda.

Frame Future Work in Terms of PhD Thesis Topics. A heuristic I like to follow is to list future directions that can be compelling thesis topics. The supporting sentences in the paragraph on each future direction might then point to specific results (e.g., specific papers you might write). Don't present future directions that are scoped at the scale of individual papers, as those don't really support your medium-level agenda.

Rule of Threes. This point is very much a personal preference, but I like to organize according to groupings of threes. Sometimes, this doesn't make sense, e.g., if there are only two natural groupings of your research. But I've found this to be a useful heuristic, as groups of three tend to occupy the right level of granularity to show both cohesion (not too many groups) and specificity (your agenda has details & substance). This is a good way to structure your medium-level agenda.

Present Refined Research Taste. The cumulative effect of the above points is to paint a picture of a researcher who has great research taste, who can have productive research conversations at any level of detail. We all have more problems to work on than we have time to do, so how do you choose to what to work on with your limited time?

. . .

Teaching Statement

The teaching statement is the place where you express your teaching philosophy.

Reference Statements.

- [My teaching statement](#)
- [Sarah Dean's teaching statement](#)
- [Angie Liu's teaching statement](#)
- [Guanya Shi's teaching statement](#)
- [Ziv Scully's teaching statement](#)

- [Sara Beery's teaching statement](#)
- [Yuxin Chen's teaching statement](#)
- [Anjalie Field's teaching statement](#)
- [Jennifer Sun's teaching statement](#)

State your experience, and do so to support your philosophy. It is important to state your prior experience, but to do so to support your teaching philosophy. In contrast to your research statement, I think it's fine to brag a little more about prior teaching experience, as I find it less common for recommendation letters to go into substantial detail about teaching.

Separate out classroom teaching versus advising. Both aspects are important and worth commenting on separately.

State what classes you're able to teach. For schools that have specific teaching needs, not being able to teach certain classes can really ding you. So be explicit about what classes you feel comfortable teaching.

. . .

Diversity Statement

I don't have as much advice here, as how faculty search committees evaluate diversity statements is an evolving process. Some generic tips (which are completely my nascent personal opinion).

Reference Statements.

- [Sarah Dean's diversity statement](#)
- [Guanya Shi's diversity statement](#)
- [Angie Liu's diversity statement](#)
- [Ziv Scully's diversity statement](#)
- [Sara Beery's diversity statement](#)
- [Anjalie Field's diversity statement](#)
- [Jennifer Sun's diversity statement](#)

Communicate Thoughtfulness. We recognize that many (most?) applicants have not yet have substantial experience working on issues pertaining diversity, equity, and inclusion. However, statements that demonstrate

thoughtfulness are still appreciated. Thoughtfulness is one of the most important indicators of future positive impact. To be clear, thoughtfulness basically goes hand-in-hand with educating oneself on the issues.

Communicate Dedication. While many applicants will not have had substantial experience in this space, we do look to see how applicants have made use of available opportunities. Dedication is the other important indicator of future positive impact.

Constructive Paths Forward. We recognize that not everyone agrees with all the ideas being floated around regarding DEI. Some stances & policies, if taken too far, might be counterproductive in other ways. The best way to communicate these viewpoints is to offer constructive paths forward. Nuance is key here.

Communicate Guiding Principles/Philosophy. As a professor, you may only be able to dedicate, say, 5% of your time to promoting DEI. How do you go about prioritizing your limited time? What activities do you find most impactful or important? How do we address systemic issues?

. . .

Representative Publications

Most schools ask for a list of representative publications (typically 3). You should expect reviewers to read at least one of these papers (especially if you get into the short list of applicants). You should also expect reviewers to look at the papers in order (1st representative paper, then 2nd, then 3rd).

For reference, these were the three representative papers I used for my application in 2012–2013.

Submit Your Best Work. One common misstep that many applicants make (and indeed that I made myself) is to submit the most recent papers that one is most excited about, rather than the papers that represent the strongest contributions. What can happen then is either the reviewer is not impressed because the paper seems a bit unimpressive (relative to expectations of a representative paper), or the reviewer needs to dig to find the more seminal

result of your previous research. Such a misstep is typically not a deal-breaker, but can make it harder for the reviewer to put together a holistic picture of your best moments. (Of course, sometimes your best work is your most recent work.)

Strike a Balance of Coverage vs Focus. On the one hand, you don't want to submit papers that are redundant (e.g., a series of results that are from a linear research thread). On the other hand, it can be risky to submit preliminary results that you've only been recently exploring (which have the benefit of showing what directions you're interested in pursuing in the future), but the paper is kind of weak. My rule-of-thumb is that it's ok to have one such preliminary results paper in a set of 3 representative papers.

. . .

Recommendation Letters

In most situations, recommendation letters are the most important part of your application package. Recommendation letters come from established researchers in the field who can speak to your contributions (and brag about them in a way that's awkward for you to do so), and place them in context with respect to the other work happening in your field. While you don't have direct control over the contents in the letters, you do have some things you can influence.

Avoid Dilution. Generically speaking, I think the maximum number of letters should be five. The most healthy number is usually four. You don't want a reviewer to be left with the impression that half your letter writers think you're simply "OK" if the goal is to shine through compared to other applicants.

Ask Letter Writers to Comment on Non-Standard Circumstances. If you had a non-traditional pathway or experience during your PhD and/or postdoc, it often goes over better if a letter writer is able to discuss those issues objectively. Issues can include, but are not limited to, dealing with disabilities, personal leave, unusual set-backs in research such as needing to switch areas, and poor relationships with previous advisors. Of course, it's important to have a frank discussion with your letter writer to determine

whether it's helpful to comment on these aspects. But the first step is to bring these issues up for a discussion.

Writing Letters is the Commitment, not Submitting Letters. It takes more time to write a recommendation letter than to submit that letter to 30 places. If you're asking someone to write a letter of recommendation for you, prefacing it with "it's just for a couple of schools" doesn't actually help. In some cases, it can actually create a negative feeling on the part of the letter writer in that they're doing all this work only to submit the letter to 2–3 places.

Give Letter Writers Enough Time. Most letter writers prefer having a few weeks notice (unless they already have a previously written letter that they can re-use). It takes time to write a thoughtful recommendation letter.

Faculty Application

Tips

**Written by Yisong Yue**

517 followers · 203 following

Machine Learning Professor @Caltech

Follow

Responses (2)



Write a response

What are your thoughts?

**Keqiang Yan**

Jul 3, 2024

